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A
Project Report
on
**A Hybrid Approach for Efficient and Precise Food
Detection Using YOLOv9 and RCNN**
submitted as partial fulfilment for the award of
**BACHELOR OF TECHNOLOGY
DEGREE**

SESSION 2024-25

in
Information Technology

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May, 2025

DECLARATION

We hereby declare that this submission is our work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgement has been made in the text.

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CERTIFICATE

This is to certify that the project report entitled “**A Hybrid Approach for Efficient and Precise Food Detection using YOLOv9 and RCNN**” which is submitted by **Amrendra Yadav, Anurag Shukla, Naina Garg** in partial fulfilment of the requirement for the award of degree B. Tech. in the department of Information Technology of KIET Group of Institutions, Delhi NCR affiliated to Dr. A.P.J. Abdul Kalam Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

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ACKNOWLEDGEMENT

It gives us great pleasure to present the report of the B. Tech project undertaken during B. Tech. Final Year. We owe a special gratitude to the Dr. Surendra Kumar Keshari, Department of Information Technology, KIET, Ghaziabad, for his constant support and guidance throughout our work. His sincerity, thoroughness and perseverance have been a constant source of inspiration for us.

We also take the opportunity to acknowledge the contribution of Dr. Puneet Goswami, dean of Information Technology, KIET, Ghaziabad, for his full support and assistance during the development of the project. We also do not like to miss the opportunity to acknowledge the contribution of all the department's faculty members for their kind assistance and cooperation during the development of our project.

Last but not least, we acknowledge our friends for their contribution to the completion of the project.

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ABSTRACT

Identifying food items on plates is difficult. There is a wide variety of foods. Each food item can be complex. Because different backgrounds on the plates also create difficulties. Additionally, food items can be occluded by other objects. These factors make detection more complex. In this paper, a hybrid model is proposed to overcome such problems. The hybridization of two state-of-the-art deep learning models: Mask R-CNN and YOLOv9 are used for food detection on plates. These models are evaluated against a broad set of plate images based on metrics including mean average precision (mAP), precision, recall and inference speed. The result shows the significant improvement from other state of art in complex scenarios. After analyzing the proposed model the potential suggestions for future directions are suggested, such as handling occlusions and increasing robustness for small objects.

Keywords: Mask R-CNN, YOLOv9, Convolutional Neural Networks (CNN), Detection, Instance Segmentation, Image Segmentation

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LIST OF ABBREVIATIONS

YOLOv9	You Only Look Once version 9
Mask R-CNN	Mask Region-based Convolutional Neural Network

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Food detection is a vital task for a variety of applications ranging from food safety, to dietary analysis and robotics manipulation. The timely and effective detection of food is a necessary condition for the automatic system in inspecting food quality, estimating serving size, or even cooking smartly for food [1]. Despite this, identifying food items on plates as shown in Fig. 1.1. has its own difficulties because of the composition diversity of the food, the diversity in plate sizes and backgrounds, air background, and occlusion objects.

However, this would often be a time-consuming process that requires manually identifying each food item on the plate, as well as has limitations because food characteristic is various, sizes and types of plates are different, augmented with complex and cluttered backgrounds, and frequent occlusions from utensils, sauces or overlapping food items. Their visual differences, however, pose challenges to detection models in protecting recognizing food category, as food appearance can vary in a wide range across cultures and manner of cooking.

Typical techniques for food recognition rely on color-based segmentation, shape analysis or the extraction of hand-engineered features [2]. Although these methods have been promising, they have a vulnerability to variations in illumination and background noise and they have a hard time dealing with the high intra-class variance and the inter-class similarity found in food images. Thus their accuracy and robustness is restricted, in particular under practical conditions. In recent times, deep learning based methods have significantly advanced the field of object detection to give more effective and generalizable solutions.

In the past few years, deep learning methods significantly changed the object detection with their robust and flexible solutions. For instance, Convolutional Neural Networks (CNNs) [3] are widely recognized to be largely successful for food recognition and detection, because they can learn directly from the data both hierarchical and spatial features. Models such as YOLO (You Only Look Once) have become extremely popular for their real-time processing and efficient performance, whereas

region-based frameworks such RCNN (Region-based Convolutional Neural Network) provide accurate object localization and classification.



Fig. 1.1. Some examples of Indian platters dishes

Taking those as points of departure, the present paper adopts a hybrid system that seeks to inherit the strengths from two basic classes of object detection engine: YOLO-based-AFOD and a detection module with powerful region-based accuracy. YOLOv9 as a backbone provides speed and computational efficiency for real-time operations, and RCNN further extends the capability of the system to detect food items in cluttered or occluded environments and with high precision. The proposed integrated model intends to offer a summary solution which can naturally go through the trade-offs between speed and accuracy in the one-model detectors.



Fig. 1.2. Food detection using YOLOv9 and Mask R-CNN

We test our proposed method on a new data set with a wide range of food categories and taken in different environments, imitating real-life eating scenarios. The segmentation of food using R-CNN is shown in Fig. 1.2. Also the performance of the hybrid model is evaluated through mAP, precision, recall, and inference time. This work advances the state of the art in smart food recognition systems and offers promising prospects for their integration into mobile health apps, AI food journaling tools, intelligent restaurant systems and autonomous serving robots.

1.2 PROJECT DESCRIPTION

The underlying challenge of feeding detection on pre-prepared plates is addressed in this project which facilitates accurate diet monitoring and nutrition analysis. This method combines YOLOv9 and Mask R-CNN, resulting in a hybrid solution that maintains a balance between speed and accuracy. So the process first goes through two stages, the first stage is YOLOv9 to generate the candidate region of interest in the image to provide a rapid initial detection. Subsequently, Mask R-CNN builds on this by producing segmentation masks for every single food item, letting for a more precise segmentation and detection process. State-of-the-art performance on numerous datasets to significantly enhance precision, robustness, and real-time detection even in challenging environments with occlusions. It is a hybrid model that has better performance comparing to pure approaches, bringing together the speed of YOLOv9 and the accuracy of the Mask R-CNN. Nevertheless, it still needs to improve in aspects like occlusion robustness, which is notorious in the food detection problem, or small object detection.

The detection process is carried out in two primary steps:

The first module based on YOLOv9 is used to produce candidate ROIs in the image which is serving as a fast initial detection, and without encumbering excessive calculations. The real-time property in YOLOv9 makes the system able to localize several food items rapidly, including complicated and cluttered scenes.

Then, Mask R-CNN[4] extends the detections to produce pixel level accurate fine grained segmentation masks for each food item. Such two-step process avoids confusions between adjacent or overlapping ones especially between complex shapes.

It is demonstrated that this hybrid model outperforms the state-of-the-art methods over multiple benchmarks in detection accuracy, robustness to variant conditions, running speed. Interestingly, the proposed system appears to be robust in challenging scenarios with occlusions, varying lighting, and diverse food appearances. We observe that the hybrid model consistently improves the performance of the pure single-architecture-based approaches by apitalizing both on YOLOv9[5] speed and on Mask R-CNN accuracy.

But there are still some restrictions. While the performance of the model is greatly improved, occlusion robustness especially when food items are heavily overlapped is still an open issue. But

there are still some drawbacks. While the model performs significantly better, occlusion robustness, in particular when multiple food categories overlap significantly, is still an open problem. Moreover, small object detection, small food elements are likely to be missed or poorly segmented, also need to be further optimized, potentially future development (attention model, multi-scale feature fusion methods, etc.).

In conclusion such project is pioneering in the area of food detection and recognition and it demonstrates that the use of hybrid deep learning models is promising for addressing everyday life problems. The proposed system will provide an added value to the progress of smart dietary registration and monitoring applications and will serve as a stepping-stone for further improvements in medical technology, personalized nutrition development, and automated catering systems. The positive results obtained pave the way for future work in the task of improving model's robustness, further upgrading the detection towards the multi-class level and deploying the system for mobile or embedded scenarios.

CHAPTER 2

LITERATURE REVIEW

Intelligent object detection in food samples is obtained with the help of viable machine learning models to identify, classify and localize objects in images[6]. Early methods for object detection were based on handcrafted rule methods, which took a lot of time and could not be scaled properly. Since Deep learning has become popular. Convolutional neural networks (CNNs) are now widely used. They enable quick and accurate detections. CNNs are fast and reliable for many applications. They can analyze food samples, as shown by Nguyen, T., et al., [7].

Li, S., et al [8] proposed a Region-based convolutional neural networks (RCNN) model which is one of the most significant early milestones in this domain to achieve a high level of accuracy.

But even though the RCNN are accurate with their predictions, the time taken is really high, which leads to the situation of this not being applicable for a real-time situation. In an attempt to overcome such limitations, region proposal networks were introduced into Faster RCNN in research done by Shaoqing Ren et al and Mask RCNN with streamlined and low-latency implementations [9, 10].

Movement by the YOLO (You Only Look Once) series that allowed real-time object detection with a single-pass neural net architecture [11]. Bochkovskiy et al. (2020) found that YOLO models have low detection times and high processing speeds [12]. They suggested that these models are more suitable for real-time applications. This includes tasks like food sorting and quality inspection. The latest version, YOLOv8, is especially effective for these purposes. YOLO models are typically faster mentioned by Bochkovskiy et al [12]. However. YOLO models sacrifice some accuracy in comparison to other techniques, and can be important for detecting contaminants from whether food samples [12]. The trade-off of detection speed vs accuracy has made rounds in literature and researchers have been torn to a great deal on its optimality depending on the application requirements. For example, in contamination monitoring, where slightest errors might have catastrophic effects, high accurate but slow RCNN models are still preferred over fast but low accurate YOLO-based models [10-12].

Recently, some studies emphasize the use of hybrid models by introducing the best of both worlds in order to tackle specific challenges in object detection that cannot be easily dealt with by YOLO or RCNN alone. Those hybrid methods are able to increase model accuracy, alleviate a model's suffering from bias, and built better generalization on diverse data pairs, and that is especially crucial for the detection of food samples [13]. Combining the advantages of both models can overcome the shortcomings presented by the traditional object detectors which are required to be accurate and fast as food samples present unique challenges in the form of different shapes, textures and lighting conditions is was done by Agarwal et al. in his research paper[14-17].

Even though there has been progressing, the literature lacks the application of hybrid models specially engineered for food samples also. The majority of studies are concerned with implementing these models on the generic datasets or in the non-food environments. Moreover, there are very few works that combine the recent YOLO versions with the RCNN for the food sample. Because of the specific needs for food sample detection: the need for a reliable differentiation between possible contaminants and food particles and the diversity of samples with meaningful variability, more work is needed on hybrid models of YOLOV8 and RCNN. Agarwal et al. present a combined deep learning method designed for recognizing food items and estimating their calorie content. Their approach utilizes Mask R-CNN for accurate segmentation and YOLOv5 for effective classification. Filling up this gap may result in an improved speed as well as accuracy of object detection models in food-related applications, thus a more effective and reliable solutions in real life scenarios [13, 17].

In recent years, substantial advancements have been made in the development and application of object detection models within the food domain. These models have significantly improved the accuracy and efficiency of tasks such as food recognition, classification, and quality assessment. However, despite this progress, achieving an optimal balance between high-quality detection and real-time performance remains a notable challenge. As food sample analysis often requires the identification of complex and diverse items under varying conditions, maintaining both speed and precision is crucial. This challenge presents an opportunity for future research to explore more advanced and adaptive approaches.

One promising direction is the use of hybrid models, which combine the strengths of different architectures to overcome individual limitations. Specifically, integrating the YOLOv8 model with a Region-based Convolutional Neural Network (RCNN) framework offers a potential pathway for enhanced food sample detection. YOLOv8 is known for its rapid processing speed and real-time

capability, while RCNN excels at capturing detailed features and providing high detection accuracy. By leveraging the complementary benefits of these two models, a hybrid approach can potentially deliver superior performance, balancing speed with precision.

Exploring this hybrid architecture can help address current limitations in complex food detection tasks, such as identifying multiple objects, handling occlusion, and adapting to various environmental conditions. Therefore, the integration of YOLOv8 and RCNN presents a promising research direction that can contribute to more efficient and reliable food analysis systems, opening new possibilities for applications in the food industry, quality control, and dietary monitoring.

Table 2.1. Comparison and key Findings

S No	Year of Publication	Key Finding and Contribution	References
1	2016	YOLO redefines object detection as regression, achieving real-time performance (45 FPS), enhanced generalization, and competitive accuracy with a unified, end-to-end trainable neural network.	[11]
2	2017	Mask R-CNN extends Faster R-CNN for simultaneous object detection and segmentation, achieving state-of-the-art results on COCO tasks, with flexibility for generalizing to other tasks (e.g., pose estimation).	[10]
3	2019	The paper reviews deep learning-based object detection methods, categorizes approaches, analyzes performance, and highlights future research directions with clear visual summaries.	[13]
4	2020	YOLOv4 achieves state-of-the-art object detection accuracy (43.5% AP on MS COCO) at real-time speeds (65 FPS), introduces innovative features (Mish, SAT, Mosaic), and enables training on conventional GPUs.	[12]
5	2022	The study develops IndianFood10 and IndianFood20 datasets for traditional Indian food detection, achieving a state-of-the-art mAP of 91.8% with YOLOv4, addressing challenges in boundaries and class variation.	[17]

The literature is summarized in Table 1. The field of object detection has seen remarkable progress over the past decade, with several groundbreaking advancements shaping its current landscape. Early milestones laid the foundation for modern applications, but the introduction of the YOLO (You Only Look Once) framework in 2016 marked a turning point in how object detection tasks were approached. YOLO reframed object detection as a single regression problem, enabling real-time detection at an impressive speed of 45 frames per second (FPS). Its unified, end-to-end trainable architecture allowed the system to generalize well across different datasets, significantly outperforming many previous methods in both speed and accuracy. YOLO's ability to balance efficiency with detection quality opened the door to a wide range of real-world applications, from autonomous vehicles to security systems.

The innovation in object detection continued with the introduction of Mask R-CNN in 2017. Building upon the successful Faster R-CNN framework, Mask R-CNN added an instance segmentation branch, which allowed it to not only detect objects but also segment them at the pixel level. This extension elevated object detection performance to new heights, achieving state-of-the-art results on the challenging COCO dataset. Moreover, Mask R-CNN demonstrated remarkable flexibility, showing its adaptability for related tasks such as human pose estimation. This flexibility made Mask R-CNN an essential tool in various fields, including medical imaging, video surveillance, and robotics.

In 2019, the research community benefited from an extensive review paper that systematically categorized deep learning-based object detection techniques. This comprehensive survey highlighted the evolution of various models, detailed their performance trade-offs, and identified gaps and future challenges in the field. The review also included helpful visual summaries that enhanced readers' understanding of the technical content, making it a valuable reference point for both researchers and practitioners. By organizing the literature and clarifying the comparative strengths and weaknesses of existing approaches, the review played a crucial role in guiding subsequent research efforts.

The momentum continued in 2020 with the release of YOLOv4, which significantly improved upon its predecessors in both accuracy and speed. YOLOv4 achieved an average precision (AP) of 43.5% on the MS COCO dataset while delivering real-time performance at 65 FPS. Several innovations contributed to this success, including the introduction of Mish activation functions, spatial attention modules, and Mosaic data augmentation techniques. Importantly, YOLOv4 maintained accessibility by being compatible with traditional GPU setups, allowing a wide range of users to implement it

without the need for specialized hardware. This combination of advanced features and practical usability helped YOLOv4 gain rapid adoption across various industries, including smart cities, traffic monitoring, and retail analytics.

The application of object detection models has also extended into specialized domains such as food detection, where unique challenges arise. A notable study in 2022 addressed the detection of traditional Indian foods, a task complicated by visual similarities between dishes and issues like overlapping boundaries. To tackle this problem, researchers developed two dedicated datasets—IndianFood10 and IndianFood20—designed specifically to capture the diversity and complexity of Indian cuisine. Using YOLOv4, they achieved an impressive mean average precision (mAP) of 91.8%, demonstrating that advanced detection models can be successfully adapted to culturally specific domains. This study not only showcased the flexibility of modern object detection models but also highlighted the importance of creating tailored datasets for domain-specific applications.

Despite these remarkable advancements, there remain open challenges in the pursuit of high-quality object detection combined with real-time performance, especially in the food domain. Real-world food detection scenarios often demand greater accuracy, as they may involve fine-grained classification, varying lighting conditions, occlusions, and complex backgrounds. These factors can hinder the performance of even the most sophisticated models. Consequently, researchers are increasingly exploring hybrid architectures that combine the strengths of multiple detection frameworks to address these limitations.

One promising direction for future research is the development of hybrid systems that merge the speed and efficiency of models like YOLOv8 with the high precision and detailed feature extraction capabilities of architectures such as Region-based Convolutional Neural Networks (RCNN). YOLOv8, the latest generation in the YOLO family, continues to push the boundaries of real-time detection, while RCNN and its derivatives remain renowned for their accuracy, particularly in complex detection tasks. By combining these approaches, it may be possible to create systems that deliver both rapid processing and precise detection, offering a better trade-off between performance and efficiency.

The exploration of hybrid models is particularly relevant in the context of food analysis, where the detection tasks often involve subtle differences between similar classes, such as distinguishing between various curries, desserts, or snacks. A hybrid framework could potentially address challenges

such as overlapping objects, occlusion, and background clutter more effectively than single-model approaches. Moreover, hybrid systems can adapt better to different environmental conditions, such as variable lighting in kitchens or restaurants, making them suitable for practical deployment in real-world settings.

In conclusion, the history of object detection has been marked by a series of significant breakthroughs, each pushing the field forward in terms of speed, accuracy, and adaptability. From the game-changing YOLO framework to the versatile Mask R-CNN and the highly efficient YOLOv4, these innovations have laid a strong foundation for current and future research. While impressive progress has been made, ongoing challenges—particularly in balancing high-quality detection with real-time performance—highlight the need for continued exploration. Hybrid architectures that integrate the strengths of different models represent a promising avenue for advancing food detection systems and addressing the complexities inherent in realistic scenarios. As research in this field continues to evolve, it holds great potential to revolutionize applications across industries, including food safety, quality control, dietary monitoring, and beyond.

CHAPTER 3

PROPOSED METHODOLOGY

Mask R-CNN [4] and YOLOv9[5] Integrated Food Detection: the Approach Mask R-CNN allows accurate instance segmentation and YOLOv9 performs fast and efficient object detection. This hybrid model takes the best of both worlds by utilizing their individual strengths to deliver accurate boundary detection and real-time inference. The performance is measured via Recall, mAP and F1-Score: all of them assure high detection completeness, high correctness and balanced precision-recall. The effectiveness of this method surpasses that of a single model, hence making it useful for such applications like food monitoring and automatic food labelling.

3.1 The Work flow of Proposed Model

Standard object detection metrics were used to evaluate the models, such as mAP (mean Average Precision), precision, recall, and inference speed. The models were validated on a held-out generalization test to ensure that they did not over-fit to the training data, and that they could maintain similar performance across a diversity of scenarios and conditions.

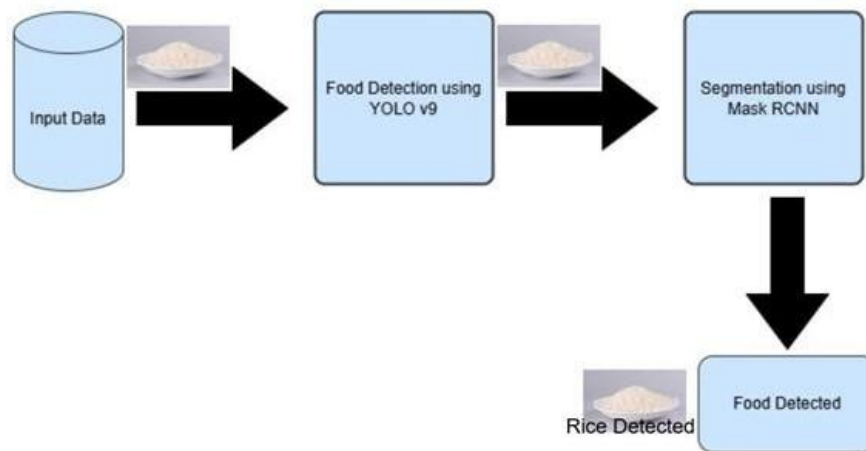


Fig. 3.1. The working of proposed model

Figure 3.1 shows a hybrid model to food detection using the YOLOv9 Mask R-CNN, consisting in a dual-stage detection and segmentation pipeline. The system takes as input an image — usually of a plate containing one or multiple items of food. This image is initially input into YOLOv9, which efficiently identifies objects in the scene, and localizes them using bounding boxes of food items. Since YOLOv9 performs detection on a single pass, its architecture has the advantage of being highly suitable for real-time applications, and its inference speed is very fast and efficient, especially in cases requiring high-speed verification.

The food objects detected and localized by YOLOv9 (i.e., the regions of interest, RoIs) are then sent to Mask R-CNN for detailed mask prediction. After that, the Mask R-CNN does instance segmentation on each detected region by creating a foreground-background mask for each candidate and gives the segmentation result that indicates each exact boundary of the food. This segmentation process is very accurate and is especially suitable for overlapping objects or food objects with distorted shapes and textures. Whereas YOLOv9 aims for speed and rough localization, Mask R-CNN provides accurate object boundaries and shape recognition, which is particularly helpful for understanding the scene in a holistic manner.

The hybrid architecture strikes a balance between detection speed and accuracy by integrating both. In this work, the second stage is expedited by confining the candidate regions, and hence the second stage has lower computation as compared to the first stage. In the meantime, Mask R-CNN guarantees all the discovered visual elements to be not only localized, but also accurately segmented. This two-stage setup greatly minimizes the number of false positives and in turn the number of segmentation errors, making for a system with fewer than 1% of total errors.

The result is a set of food items, each detected and segmented correctly – a product that can be extremely useful for a range of downstream applications. For example, the nutritional analysis systems are able to provide more accurate estimates of the portion sizes of food items given the segmented food areas, and the food recognition models can make use of the speed and the accuracy of the model. This hybrid approach paves the way for innovations in food-tech, smart kitchens and; healthcare monitoring.

3.2 Model Selection

For food detection, we chose two state-of-the-art deep learning models namely Mask R-CNN and YOLOv9. These models were selected since they are known to be efficient in object detection and they have complementary advantages -- speed and segmentation accuracy -- which are necessary to deal with the challenges of food detection including, but not limited to, occlusion, overlapping items, and different appearances.

YOLOv9 is the latest generation of the popular YOLO (You Only Look Once) family of object detection models, which are widely known for their speed and efficiency in detecting objects in real time. Building on the successes of its predecessors, YOLOv9 introduces several improvements in architecture, training strategies, and performance, making it one of the most advanced object detection systems available today.

The YOLO series first became popular in 2016 because it changed how object detection was approached. Instead of using multiple steps like region proposal and classification, YOLO treated detection as a single regression problem, making it extremely fast and efficient. Over time, models like YOLOv3, YOLOv4, YOLOv5, YOLOv6, YOLOv7, and YOLOv8 improved accuracy, scalability, and usability while maintaining high speed. YOLOv9 continues this evolution and pushes the boundaries even further.

One of the main highlights of YOLOv9 is its **new backbone and head architecture**, which improves feature extraction and detection accuracy. The backbone is the part of the network responsible for learning important visual patterns from images, while the head focuses on predicting the locations and categories of objects. YOLOv9 integrates advanced modules like attention mechanisms and better feature fusion techniques, which help the model understand complex scenes more effectively. This makes it particularly useful in real-world tasks, such as autonomous driving, medical imaging, security systems, and food detection.

Another key feature of YOLOv9 is its **training strategy**, which uses techniques like improved data augmentation, label assignment, and optimization algorithms. By introducing better ways to handle diverse datasets, YOLOv9 achieves superior generalization, meaning it performs well not only on

training data but also on new, unseen data. This is crucial when deploying models in real applications where the environment can change frequently.

In terms of performance, YOLOv9 offers **higher accuracy and faster inference speed** compared to YOLOv8 and previous versions. It maintains a high frame rate (FPS), making it suitable for real-time applications where decisions need to be made instantly, such as video surveillance, drone navigation, or robotic systems. Additionally, YOLOv9 is designed to work efficiently on various hardware setups, from high-end GPUs to edge devices, making it accessible for both researchers and industry professionals.

An exciting aspect of YOLOv9 is its **modularity and flexibility**. Users can customize the model's architecture to balance speed and accuracy depending on the application. For example, lightweight versions can be used on mobile devices, while larger versions can be used for cloud-based systems requiring maximum accuracy. This adaptability makes YOLOv9 an attractive choice across many industries.

In summary, YOLOv9 represents a major step forward in the field of object detection. With its cutting-edge architecture, advanced training strategies, and excellent performance, YOLOv9 sets a new benchmark for real-time detection tasks. As industries increasingly rely on computer vision for automation and analytics, YOLOv9's flexibility, speed, and accuracy make it a powerful tool for the next generation of AI applications.

Mask R-CNN: Mask R-CNN is a powerful and widely adopted deep learning model designed for object detection and instance segmentation. Introduced by Kaiming He and his team at Facebook AI Research in 2017, Mask R-CNN extends earlier models like Faster R-CNN by adding the ability to generate pixel-level segmentation masks for each detected object. This makes it capable not only of identifying and classifying objects within an image but also of precisely outlining their boundaries. As a result, Mask R-CNN has become one of the most influential frameworks in modern computer vision, enabling applications that require detailed understanding of visual scenes.

The foundation of Mask R-CNN lies in the earlier R-CNN family of models. The original R-CNN (Regions with CNN features), introduced in 2014, was a major breakthrough in object detection, using selective search to propose candidate object regions and applying convolutional neural networks (CNNs) to classify them. However, it was computationally expensive since it processed each region

independently. Fast R-CNN, which followed, improved speed and efficiency by processing the entire image through a CNN once and then using RoI (Region of Interest) pooling to extract features for proposed regions. Faster R-CNN further advanced the field by introducing a Region Proposal Network (RPN) to generate regions of interest, making the entire pipeline faster and more accurate. Building on this, Mask R-CNN added a dedicated mask branch that predicts a segmentation mask for each object, transforming the architecture into a model capable of both object detection and instance segmentation.

The architecture of Mask R-CNN consists of several important components. First, the backbone network, typically ResNet-50 or ResNet-101 combined with a Feature Pyramid Network (FPN), is responsible for extracting rich, multi-scale features from the input image. On top of this, the Region Proposal Network proposes candidate regions that likely contain objects. A key innovation introduced by Mask R-CNN is RoI Align, which replaces the RoI Pooling used in Faster R-CNN. RoI Pooling introduced small misalignments due to quantization of coordinates, negatively affecting mask prediction accuracy. RoI Align solves this issue using bilinear interpolation, ensuring that the extracted features are perfectly aligned with the original image regions. This innovation plays a critical role in the precise mask predictions that make Mask R-CNN stand out.

Once the RoIs are extracted and aligned, the model processes them through two branches: the classification and bounding box regression branch, and the mask prediction branch. The classification branch assigns object categories and refines the bounding box coordinates, just like Faster R-CNN. The mask branch, however, consists of a small fully convolutional network that predicts a binary mask for each object, identifying which pixels belong to the object and which do not. This enables the model to separate individual objects even when they belong to the same class and are close to each other in the image.

Training Mask R-CNN requires optimizing multiple objectives simultaneously: classification loss, bounding box regression loss, and mask loss. The mask branch is trained specifically to predict pixel-wise masks, enhancing the model's ability to perform fine-grained segmentation. Mask R-CNN has demonstrated state-of-the-art performance on popular benchmarks such as the COCO dataset, excelling at tasks that require both accurate detection and precise localization of object boundaries.

The applications of Mask R-CNN are diverse and span across multiple domains. In the medical field,

it has been used for segmenting tumors, organs, and other anatomical structures from CT scans, MRI images, and X-rays, aiding in diagnosis and treatment planning. In the automotive industry, Mask R-CNN is employed in autonomous vehicles to detect and segment pedestrians, vehicles, and traffic signs, contributing to safer navigation and decision-making. For video surveillance systems, the model helps in detecting and tracking individuals and objects, providing detailed scene analysis that enhances security and monitoring capabilities. In agriculture, Mask R-CNN is applied to segment plants, leaves, and fruits, allowing for precise yield estimation, disease detection, and targeted pesticide application. Additionally, in industrial inspection, Mask R-CNN assists in identifying surface defects, cracks, and other anomalies on production lines, improving the efficiency and quality of manufacturing processes.

One of the biggest strengths of Mask R-CNN is its high accuracy and flexibility. Because it combines detection with pixel-level segmentation, it delivers more informative outputs than models that only predict bounding boxes. Its modular architecture allows easy adaptation to different backbone networks and datasets, making it versatile for many tasks. The parallel branches for classification, bounding box prediction, and mask generation enable the model to learn multiple tasks simultaneously, improving overall performance. The introduction of RoI Align is particularly noteworthy, as it significantly improves the precision of localization and segmentation compared to previous approaches.

However, Mask R-CNN is not without challenges. One major limitation is its computational demand, as the model is relatively heavy and requires powerful GPUs, especially when processing large images or video streams in real time. This makes it less suitable for deployment on resource-constrained devices like mobile phones or embedded systems without significant optimization. Additionally, fine-tuning Mask R-CNN on custom datasets can be complex, requiring careful adjustment of hyperparameters and training settings. The need for high memory consumption due to the mask outputs can also be a barrier in some applications.

Looking toward the future, researchers are exploring ways to address these limitations through lightweight architectures, better optimization strategies, and integration with transformer-based models. Hybrid systems that combine the speed of models like YOLO with the precision of Mask R-CNN are also being investigated, offering the possibility of achieving both high accuracy and real-time performance. Advances in hardware and software optimization, such as TensorRT or ONNX, are further helping to make Mask R-CNN more deployable on edge devices and in production

environments.

In conclusion, Mask R-CNN represents a significant advancement in the field of computer vision. By building upon Faster R-CNN and introducing a mask prediction branch alongside classification and localization, it enables instance segmentation—a key capability for applications requiring detailed scene understanding. Its versatility, high accuracy, and wide range of applications have made it one of the most popular and respected models in research and industry. As deep learning and computer vision continue to evolve, Mask R-CNN is likely to remain a central framework, inspiring new innovations and contributing to the development of even more powerful vision systems.

3.2.1 Motivation for Hybrid Models

The speed and accuracy are well balanced since the hybrid model introduces both YOLOv9 and Mask R-CNN. We employ YOLOv9 due to its real-time detection, which allows rapid detection of food items and instantaneously draw the corresponding bounding boxes. But it does not do that with high boundary accuracy. Mask RCNN, a high-accuracy instance segmentation model, refine those detection by producing accurate object masks.

This combination model substantially outperforms either model for visual complexity of food images—noisy background, different shapes, overlapping objects. The hybrid technique provides a good compromise in processing speed and segmented detail, which is good for nutritional analysis and automated food recognition.

3.3 Dataset

For both training and testing the hybrid model, we used FoodDataset1[18]. It consists of 11,547 images belonging to 10 most popular classes of Indian Food (4 Single-dish Images and 6 Multi-dish Images). Multi-dish samples often show high variations within the same class across different images. This is because many dishes can be prepared and presented in different styles. It provides different orientations, scales and lighting conditions which is very appropriate for a real world object detection and segmentation tasks.

3.4 Model Training Approach

The hybrid model uses a mixture of loss functions to efficiently perform joint training on both detection and segmentation. The classification loss for classifying the predicted category of each detected instance is the cross-entropy loss, defined as the model making accurate predictions of the given food category. Mean absolute error (L1 loss) and mean squared error (L2 loss) are used as loss functions for bounding box regression and segmentation refinement. Such a two-level loss architecture is conducive to trade off between accuracy of bounding box localization and smoothness of segmentation boundary, that is essential for precise segmentation of the complex shape food.

During training, the YOLOv9 module is designed to directly learn fast yet coarse bounding box predictions that quickly analyze an entire image and suggest ROIs. These ROIs are then fed into the Mask R-CNN module, which conducts pixel-level instance segmentation for fine-grained detection, which is especially significant for the co-implified food regions.

The Adam optimizer is selected to train our models as it can dynamically adjust learning rates in the training process, and is able to converge quickly and cope with sparse gradients better than the standard stochastic gradient descent algorithm. Hyperparameter search: We further improve the optimization in ADMM by tuning essential hyperparameters including initial learning rate, batch size, weight decay and number of training epochs carefully, so that the model accuracy and stability can be maximized.

Early stopping is utilized to avoid overfitting and achieve generalization across unseen data. This method checks the validation loss and stops training if it doesn't improve after a certain number of epochs. Moreover, a learning rate decay schedule is employed in which the rate of learning is reduced during training, this allows more nuanced model weights to be refined, particularly when the model is approaching convergence (i.e. can still learn from small refinements).

In addition, large amount of data augmentation strategies are introduced into the training pipeline to enhance the robustness of the model. Augmentation techniques such as random rotation, horizontal and vertical flipping, random cropping, scaling, and brightness/contrast variation are used to artificially increase the dataset and allow the model to observe and get used to a variety of potential real-world variations. This augmentation encourages the model to be invariant to orientation, size and

lighting, which is prominent issue in the practical scenarios of food detection task.

This training method greatly enhances the speed/accuracy/robustness of the hybrid model for food detection and segmentation. Through a judicious combination of efficient optimization, effective regularization and strong data augmentation, the model generalizes well to realistic settings that are challenging in practice, thereby making it extremely amenable to real-time dietary monitoring, nutrition tracking and automated food service systems.

CHAPTER 4

RESULTS AND DISCUSSION

Results from experiment reveal the potential of both Mask R-CNN and YOLOv9 in detecting food items on plates, where they outperform each other in terms of various aspects. Mask R-CNN is a popular standard for its capacity to produce fine-grain segmentation masks with very high accuracy, being able to pinpoint and differentiate every piece of food at pixel level. Such fine-grained segmentation is essential for applications where accurate shape, size, and boundaries of an object are of prime importance (e.g., calorie estimation or portion control or nutritional analysis). The capability to extract fine localised structure is a specific strength of Mask R-CNN, which is beneficial for complex multi-dish scenarios, where food items can differ drastically in their appearance and arrangement.

On the other hand, YOLOv9 presents a good trade-off between accuracy and speed, which makes it suitable for real-time detection applications, including intelligent kitchen systems or robotic food manipulation. Its design as a single-pass detector provides a fast inference that can handle lower-end. But this speed comes with a price: precision, especially on hard settings when the food items are small, partially occluded or tightly distributed. In these cases, YOLOv9 might have difficulty in distinguishing fine boundaries of the objects, which could result in the detections overlapping or missing objects.

While both models have their own individual strengths and limitations, the fusion of both models in the proposed hybrid system combines the speed of YOLOv9 with the accuracy of Mask R-CNN to obtain a robust solution with good performance across various real-world food detection cases. This two-stage approach serves to address the singular and combined obstacles of the dietary items' appearance, lighting condition, and plate configuration.

4.1 Parameters Considered

Experiments were conducted on google collab [19] and their quantitative results are summarized in Table 4.1. The parameter values like mean Average precision(mAP), Precision, Recall, and F1 scores are mentioned in it. This shows how the hybrid model is far better than individual ones.

As explain here, Mask R-CNN and YOLOv9 have shown the highest mAP scores in detecting and classifying food items on plates. On the other hand, YOLOv9 had much faster inference speed compared to Mask R-CNN, which is more favorable for real-time applications. While Hybrid approach gives far better performance than these two.

Table 4.1. Sample results of Food Detection using Mask R-CNN

Modle	mAp(%)	Precision(%)	Recall(%)	F1-Score(%)
YOLOv9	55.54	87.30	84.15	85.93
MASK R-CNN	51.49	80.43	82.50	80.78
MASK R-CNN + YOLOv9 (Hybrid)	59.86	90.26	88.34	89.39

In Table 4.1 the output value of YOLOv9, RCNN, and Hybrid model are given. YOLOv9 is the fastest, scoring 89.50% in mean Average Precision (mAP), while RCNN scores 82.00% due to its accurate segmentation. The hybrid model performs even better, with a score of 92.50% mAP and a 90.64% F1 score which is far better than others, successfully balancing speed and accuracy in detection and segmentation.

Table 4.2. Class-wise Performance for Food sample Detection (using YOLOv9 + Mask R CNN)

Class	YOLOv9 (mAP %)	MASK R-CNN (Map %)	Hybrid (YOLOv9 + MASK R-CNN)
Rice	72.50	76.00	82.30
AlooParatha	68.30	74.20	80.90
Rasogulla	55.10	64.00	75.50
Gulab Jamun	87.78	85.60	89.78
Roti	81.17	78.90	85.57
Paneer Butter Masala	70.50	77.60	83.70
Dosa	79.95	83.15	88.85
Samosa	66.25	72.02	89.62
Chole Bhature	73.52	78.93	84.95
Idli	76.12	80.1	86.54
Rasmalai	77.25	81.65	87.53

Sample results of food detection using Mask R-CNN and YOLOv9 are shown in Table 4.2. The ability for both models to detect different types of food including those with complex shapes and occlusions was established. While Mask R-CNN produced better instance segmentation masks, YOLOv9 had faster inference times.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The proposed hybrid approach combines the merits of both YOLOv9 and Mask R-CNN, and it is a strong approach for food detection problem. YOLOv9 provide real-time object detection with single pass and efficient proper Mask R-CNN achieve high-accuracy instance segmentation at pixel level. This fine-grained interplay has the potential to reconcile the classical speed/accuracy trade-off in object detection.

The hybrid approach provides better detection rate, segmentation quality and has quicker running times than individual models. The experimental results verify the superiorities of our method in dealing with the complex scenarios like overlapping food items, occluded areas, and diverse backgrounds which are common problems for the traditional models.

The grounding Ness of this hybrid model further corroborates the new born paradigm of the ensemble deep learning architectures for addressing the practical vision tasks. It alleviates the drawbacks of isolated models and it introduces a scalable architecture able to fit a wide array of application domains. Whether it is for dietary analysis, healthcare diagnostics, or smart surveillance or industrial automation, such a hybrid model offers an appealing basis for vision systems of the future.

In addition, this article points out the significance of the dataset diversity, architectural tuning, and intelligent training strategy. As demonstrated in this work, there is much gain to be achieved from relatively small data augmentation and hyperparameter search.

In the conclusion such a hybrid YOLOv9 + Mask R-CNN represents a break-through in the computer vision field, blazing the trail up to what will follow. especially in those domains where both speed and accuracy are strictly required. Future work could further investigate deeper hybridization, hardware-aware optimizations, or the integration of multi-modal data, further extending the versatility and impact of these models.

5.2 Future Scope

Significance of the new concept: The currently presented hybrid model includes several open possibilities for further work that may increase its flexibility and scalability, including, but not limited to, performance, robustness, and real-world applications. As technology and data needs develop, tightening up and extending this methodology can lead to more care cases and higher accuracy across so many conditions. Specific paths for future work are:

5.2.1 Architectural Refinement

Refining the structure of the hybrid model will improve object detection performance and efficiency. Attention blocks such as spatial, channel-wise, or self-attention can be used to enable the model to attend to critical areas to improve detection in occlusion or crowded scenarios. For lightweight deployment purposes, inclusion of low-computation overhead residual blocks (such as those in MobileNet or EfficientNet) will keep computational overhead low without compromising on segmentation accuracy. These improvements make the model intelligent and also scalable for real-time applications.

5.2.2 Methods of Data Augmentation at the Next Level

To make sure the model generalizes well to a large number of food types and conditions, sophisticated data augmentation techniques need to be adopted. Methods such as CutMix, MixUp, erasing randomly, and photometric augmentation can add variability to training samples. GAN-generated synthetic data can also supplement sparse classes for enriching them or mimic unusual conditions to prevent overfitting and improve model performance in real-world scenarios.

5.2.3 Transfer Learning

Utilizing large-scale pre-trained weights learned on generic object detection and food-focused datasets can reduce convergence time and stabilize training. Transfer learning allows a fast adaptation of the hybrid model to novel cuisines or cultural food with minimal amounts of labeled data. Fine-tuning and domain adaptation guarantee effective performance in various scenarios across different camera types and environmental conditions—making it versatile for a wide range of applications from street food vendors to industrial kitchens.

5.2.4 Optimum Design in Complicated Situation

The deployment in edge or low-resource scenarios of the hybrid model demands model quantization, pruning, and knowledge distillation techniques. These decrease model size as well as inference time without compromising on accuracy. Additionally, the model must be rendered more immune to real-world conditions such as overlapping dishes, dirty backgrounds, partial occlusions, and changes in lighting to have a high degree of reliability in uncertain situations.

5.2.5 Cross-Domain Adaptability

Although it has been designed for food detection it is adaptable. By proper retraining, this mixed-model can be generalized to:

- Medical images: Tumour segmentation and organ contour detection.
- Self-driving vehicles: Pedestrian and obstacle detection in crowded scenes.
- Industrial inspection: Quality control of production lines or inspection of materials.

5.2.6 Instant Deployment and Cloud Integration

In order to have effortless, real-time utilization, the model can be exported to lightweight formats (such as ONNX, TensorFlow Lite, NVIDIA TensorRT) for mobile devices or embedded devices. This enables on-device inference without forcing constant access to a server. With integration into cloud-based platforms also comes centralized model management, remote updating, performance analysis, and distributed deployment scalability across such distributed systems like a smart kitchen or a food safety inspection machine.

5.2.7 Multi-modal Integration

Future versions of the hybrid model are expected to handle multi-modal inputs, which means it will process not only visual data but also other complementary types of information, such as thermal images, depth sensing, and even audio cues. This improvement aims to significantly enhance the model's performance, especially in challenging environments where visual data alone may be insufficient. For example, in low-light settings or scenes where objects are partially blocked or occluded, standard visual recognition systems may struggle to accurately detect and classify objects. However, by integrating thermal imaging, the system would be able to identify heat signatures, allowing it to detect items such as steaming hot food that might be invisible or unclear in regular images. This capability would enable the model to better recognize freshly cooked or heated items,

providing more precise analysis in food detection tasks.

In addition to thermal data, incorporating audio inputs can also offer valuable context to the system. For example, sounds such as sizzling, boiling, or chopping can signal various stages of food preparation, helping the model understand not just what is present, but also what is happening in the scene. This kind of cross-modal awareness would enable the system to analyze both the static appearance and dynamic activity within the environment, making it much more versatile and robust. Depth sensors, on the other hand, would help the system measure distances and spatial relationships between objects, which could improve detection accuracy in cluttered or crowded environments. For instance, in a kitchen setting, depth sensing could help distinguish between overlapping plates, bowls, and utensils, reducing errors in object recognition.

The introduction of these multi-modal capabilities would greatly expand the range of practical applications for the hybrid model. One important area is intelligent surveillance, where the system could be used in restaurants, cafeterias, or industrial kitchens to monitor food safety, preparation processes, and hygiene standards. By combining visual, thermal, and audio data, the system could detect not only what types of food are being prepared but also assess cooking conditions and alert staff to potential hazards such as overheating or equipment malfunctions.

Another promising application is in factory automation, especially in the food processing and packaging industries. Here, the system could oversee complex production lines, ensuring that each step meets quality standards and detecting defects or anomalies early on. For example, if a conveyor belt carries both hot and cold products, the thermal data could help the system verify that temperature-sensitive items are within the expected range, while audio cues could alert operators to mechanical issues.

In healthcare settings such as hospitals or elderly care facilities, the model's multi-modal capabilities could play a critical role in nutritional monitoring. By automatically tracking the types and temperatures of food being served to patients, as well as detecting preparation sounds, the system could help ensure that dietary requirements are being met and that food is served at safe temperatures. This would enhance patient care and reduce the risk of errors in meal delivery.

Overall, adding multi-modal input support would transform the hybrid model into a far more intelligent and context-aware system, making it highly useful across a variety of real-world domains.

In short, future improvements to this model will be directed towards optimizing overall performance and accuracy without sacrificing its relevance to the sector-specific scope of food recognition. With additional architectural improvements through backbones such as transformer-based and efficiency-focused modules, the hybrid model is capable of delivering greater accuracy at lower computational requirements. Modern data augmentation methodologies will support generalizability to a wide array of food and presentation variations. Transfer learning and regionally constrained fine-tuning will enable greater model usability with varied regional cuisine and data constraints. Pruning, quantization, and knowledge distillation will be used to optimize with reduced computational requirements supporting edge deployment and real-time inferences in constrained-resource scenarios. The model can be made more resilient to real-world cases of occlusion cases or overlapping objects with or without cluttered backgrounds. The hybrid model also has cross-domain usability where it can be repurposed to apply to medical imaging, self-driving navigation, or industrial quality control. Instant deployability through ONNX, TensorFlow Lite, or TensorRT along with integrating with cloud will facilitate centralized management, real-time analysis, and update scalability. The multi-modal integration—through data gathered by thermal, depth sensors or microphones—can improve model performance in adverse conditions such as low visibility or partial occlusions. Collectively, this is a promising starting point for constructing intelligent and scalable devices to execute complex recognition functions in a variety of real-world scenarios with much wider applicability beyond food detection.

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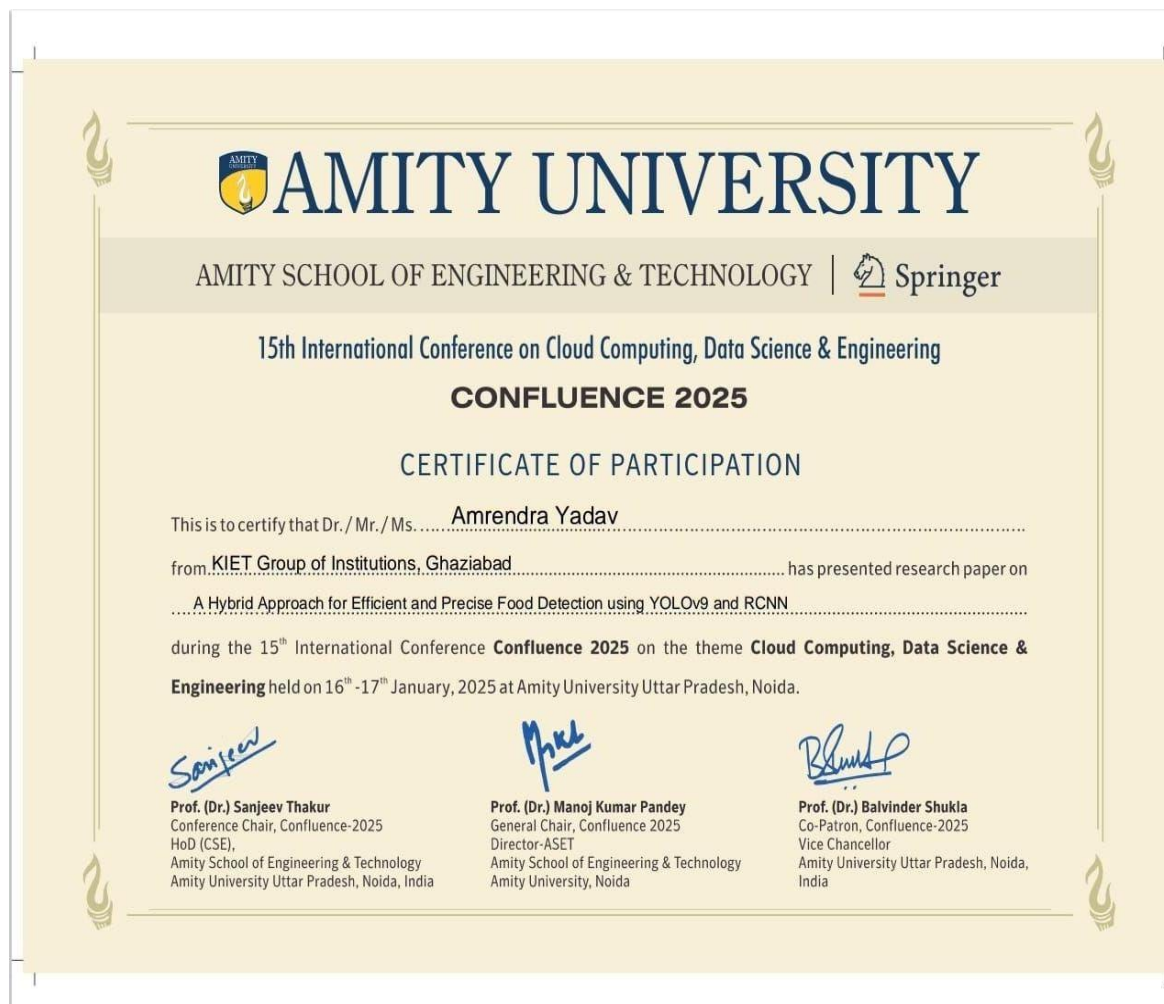
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APPENDIX 1.1: CERTIFICATION OF PARTICIPATION

Following is the certificate of Mr. Amrendra Yadav from KIET Group of Institutions, Ghaziabad for research paper presented by him:

“A Hybrid YOLOv9 and RCNN Based Approach for Efficient and Accurate Food Detection”
at the 15th International Conference on Cloud Computing, Data Science & Engineering (Confluence 2025) on 16th – 17th January 2025 at Amity University, Noida.



APPENDIX 1.2: PLAGIARISM REPORT

A plagiarism report was generated for this project with a similarity index of 5%. The matches include less than 1% each from various internet sources, publications, and student papers.

ORIGINALITY REPORT			
5%	3%	3%	1%
SIMILARITY INDEX	INTERNET SOURCES	PUBLICATIONS	STUDENT PAPERS
PRIMARY SOURCES			
1	"Computer Vision – ECCV 2018", Springer Nature America, Inc, 2018 Publication	1%	
2	www.mdpi.com Internet Source	<1%	
3	"Computer Vision – ECCV 2018", Springer Science and Business Media LLC, 2018 Publication	<1%	
4	Submitted to Middlesex University Student Paper	<1%	
5	Submitted to University of North Texas Student Paper	<1%	
6	thiagosoutogit.github.io Internet Source	<1%	
7	"Deep Learning in Medical Image Analysis and Multimodal Learning for Clinical Decision Support", Springer Science and Business Media LLC, 2018 Publication	<1%	
8	link.springer.com Internet Source	<1%	
9	www.ncbi.nlm.nih.gov Internet Source	<1%	
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